Hu Zhiheng Henrey

Software Engineer

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SUMMARY

Computer Science student at SIT and DigiPen Institute of Technology Singapore, specialising in Interactive Media and Game Development, seeking a 12 month software engineering internship. Skilled in C++, C# and Python, with hands-on experience taking software from design to a working product in teams of 5 to 8. Completed 2 team projects, each built from scratch in a custom C++ engine. Built the state management system in WAVES, and the save and load system, undo/redo, UI framework and AI behaviour logic in STRETCH. Now building a third project, a game engine application for secondary students. Communicates clearly, explaining difficult ideas simply from private tutoring, and stays calm under pressure from SCDF service. Quick to learn new tools and technologies, adaptable, and keen to explore new ideas. Aspires to build reliable, maintainable software.

EDUCATION

Singapore Institute of Technology (SIT) & DigiPen Institute of Technology Singapore Aug 2024 - Present

Bachelor of Science (Honours) in Computer Science in Interactive Media and Game Development Expected Graduation Apr 2028

Relevant Modules Data Structures, Operating Systems, Modern C++ Design Patterns, Algorithm Analysis

TECHNICAL SKILLS

Programming Languages	C++, C#, Python, JavaScript, HTML
Game Engines	Unity, Unreal Engine, Custom C++ engines
Operating Systems	Windows, Linux
Database	SQL, Chroma
Tools and Software	Git, Excel, Figma, Affinity, DaVinci Resolve

PROJECTS

Programmer | Atelier3D, 3D engine and block editor, 9-person team | C++, custom engine Sep 2026 - Present

- Beginners struggle to see how programs run. Building a 3D engine with a visual block programming game maker designed to make execution visible, as Debugging Tools Champion for the team's logging and Console tools.

Programmer | STRETCH, 2D platformer, 8-person team | C++, custom engine, RapidJSON, Unity C# Sep 2025 - Apr 2026

- The team needed progress and level data to survive crashes and restarts. Built a JSON save and load system with RapidJSON covering about 30 data types, plus auto-save and categorised crash logging so failures are traceable.
- Level designers needed to reverse mistakes in the team's in-house design tool. Solely designed and wrote an undo/redo system using JSON snapshots for edits, creation and deletion.
- The game needed menus usable on keyboard, mouse and gamepad. Developed a UI framework of 10 screens with buttons and sliders, so one codebase supports all three input types.
- Enemy behaviour needed to be proven before committing to it. Prototyped 4 behaviour types in Unity C#, then implemented them as a 4 state machine in C++. Owned enemy logic and UI as AI & UI Champion across 6 milestones.

Programmer | WAVES, 2D stealth maze game, 5-person team | C++, custom engine Jan 2025 - Apr 2025

- Five teammates' screens had to run as one application. Designed a state management system where every screen follows one lifecycle, so screens switch cleanly with no memory leaks.
- Players needed clear navigation. Built the UI flow across 8 screens and overlays, with a reusable component library and font cache.
- Led integration of four teammates' modules into one working application with end conditions.

WORK EXPERIENCE

Singapore Civil Defence Force (SCDF) | Emergency Medical Technician Apr 2022 - Feb 2024

- Responded to high-stress emergency callouts on an ambulance crew over 22 months, making real-time triage decisions.
- Worked within a rotating team, coordinating roles and communicating clearly during each callout.
- Stayed calm and reliable under repeated pressure.

Private Tutor | Geography, Social Studies, Maths and Science 2020 - Present

- Tutored 4 students since 2020 in Geography, Social Studies, Maths and Science, building their confidence in problem solving.
- Adapted lessons to each student's pace and explained difficult concepts in simple terms.
- Raised one student from a fail to B3 in Geography and Social Studies.

CERTIFICATIONS

Google Data Analytics Professional Certificate (Coursera) Jan 2026 - Oct 2026

INTERESTS

Video/Photo Editing (Affinity, DaVinci Resolve), Dizi (Grade 9), Floorball (SCDF Inter-Division Champion, 2023)